

◆ PHASE 1: User Personalization Setup

- Build a quiz UI to capture:
 - Age
 - Investment goal
 - Experience level
 - Emotional response
 - Timeline
 - Compute:
 - `risk_score`
 - `risk_tolerance`
 - Store all user data in `user_profile.json`
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◆ PHASE 2: Financial Data Scraping

- Pull news from:
 - Google Finance RSS
 - Yahoo Finance RSS
 - Stocksera (includes Reddit ticker sentiment, insider trades, lobbying, ETF flows, and politician trading data)
- Automatically deduplicate overlapping articles using:

- Title comparison
 - URL match
 - Content hashing
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◆ PHASE 3: Data Extraction & Structuring

- Use regex and spaCy to extract:
 - Percentages
 - Dollar values
 - Dates
 - Comparison phrases
 - Run VADER sentiment analysis and tag each article with:
 - compound
 - pos
 - neg
 - neu
 - Save structured output to `enriched_news_sentiment.json`
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◆ PHASE 4: Article Scoring and Clustering

- Compute an `impact_score` for each article using:
 - $|\text{sentiment}| \times (\text{keyword weight} + \text{credibility} + \text{profile match})$

- Generate sentence embeddings using `sentence-transformers`
 - Cluster articles using:
 - KMeans or Agglomerative algorithms
 - Assign `cluster_label` and select top 1–2 articles per cluster for summary
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◆ PHASE 5: GPT Summary + Validation

- Summarize selected cluster representatives using compressed GPT prompts
 - Ensure summaries retain:
 - Key financial figures
 - Quantitative references
 - Validate summaries by comparing to extracted data
 - Re-prompt if important numbers or context are missing
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◆ PHASE 6: Economic Reasoning Engine

- Upload 10–15 economic principle books (PDF format)
- Parse and embed them using FAISS or ChromaDB
- Scrape Investopedia to enrich the internal knowledge base
- Build reasoning engine to analyze cause-effect and sector impact
- Output:
 - Sector likely to be affected

- Economic logic explanation

- Risk type

◆ PHASE 6.5: Political and house

- Political lobbying and house legislature
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◆ PHASE 7: Historical Correlation Mining

- Identify similar past economic events (e.g. Fed hikes, inflation spikes)
 - Match to current market indicators
 - Output:
 - Historically high-performing sectors and stocks
 - Past performance metrics
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◆ PHASE 8: Opportunity Quantification

- Use QuantLib to compute:
 - Expected return
 - Option pricing
 - Bond yield
- Combine with:
 - Sector predictions

- Technical analysis signals
 - Generate:
 - Confidence score
 - Strategy alignment output
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PHASE 9: Backtesting Engine

- Run backtests of strategy hypotheses using Backtrader
 - Log:
 - Win rate
 - Sharpe ratio
 - Drawdown
 - Entry/exit timing
 - Portfolio exposure level
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PHASE 10: Portfolio Tracker & Dashboard

- Build mock trading engine with position tracking
- Create Streamlit dashboard with:
 - Tab 1: News Feed (sorted by impact)
 - Tab 2: Sector Insights (from reasoning module)
 - Tab 3: Strategy Results (from backtesting)

- Tab 4: Stocksera Signals and alternative data
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◆ PHASE 11: Chef GPT – Final Intelligence Layer

- Build `chef_gpt.py` to merge all inputs:
 - Summary
 - Economic reasoning
 - Quant math
 - Backtest data
 - Recession score
 - Output:
 - Final trade idea
 - Sector focus
 - Strategy type
 - Confidence level
 - Risk alignment to user profile
 - Only triggers after all other phases are complete
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◆ PHASE 12: Control Logic & Human-in-the-Loop

- Create `controller.py` to manage execution of the pipeline in proper order:
 - Scraper → Extractor → Clustering → Summarizer → Reasoner → Strategy

- Track pipeline state using:
 - `pipeline_step`
 - `execution_ready`
 - `summary_status`
 - Integrate CopilotKit to enable:
 - Manual user review
 - Optional approval or override of the final recommendation
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PHASE 13: Recession Indicator Engine (NEW)

- Build `recession_signals.py` with submodules for:
 - Macroeconomic data (yield curve, GDP, unemployment)
 - Behavioral signals (lipstick effect, fast food trends, pawn shop activity, pizza near the pentagon)
 - Social sentiment (Google Trends, Reddit signals, stress-related search spikes)
 - Market movement (defensive rotation, bond ETF inflows)
- Assign scoring weights to each category and compute:
 - Daily Recession Risk Score (0–10)
- Use score to:
 - Trigger sector alerts
 - Adjust GPT strategy recommendations
 - Display on dashboard for transparency

Also to add

Security protocols to protect backend architecture from typical vibe coded faults